Technical Documentation Blueprint for:

Water Dispensing System

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Disclaimer!

This document discusses the technical specification of the device/website that will be delivered and the items displayed are meant to be for illustrative purposes only and does not necessarily reflect the use of the same types in the device to be manufactured

Introduction

The purpose of this document is to outline the design of a smart water dispensing system. The goal of this project is to design a smart Water Dispensing System for Water Transportation trucks that can efficiently and effectively fill residential water tanks in the defined areas. The system uses the Geo-fence technology that depends on the Global Positioning System (GPS). It defines a virtual geographical boundary that used to alert the user with the truck is in near him. The system will provide clients with the amount of water allocated to them using a programmed amount that is predefined on the system website. Using the Multi-factor authentication (MFA), the system will rely on Geofencing as the first level to activate the system and will use another form of authentication as a unique account and RFID tag and/or OTP that will be used as a system user identifier key. The system is designed to be a cost-effective and environmentally friendly solution for providing clean and safe water in areas with limited access to water.

Blueprint Summary

Throughout this document we will present to the water dispensing management solution. The system will need to be able to track the trucks in different stages through set routes to track work progress and increase efficiency. i.e. is the truck enroute, on time, when will it arrive at a destination, know how much water is being dispensed into a certain unit to ensure the right amounts are being distributed to the right units, is the water clean enough to be dispensed, is it being dispensed at the correct place, reduce errors throughout the process and what is the benefit of such a system for employees, employers, and end users. This is what our solution will offer and many more.

Concept Design

The water dispensing system will be designed to be self-contained and fully automated. As shown in image 1, the truck will be equipped with a water tank, an electric valve, a water flow meter sensor, a pump, and a control unit.

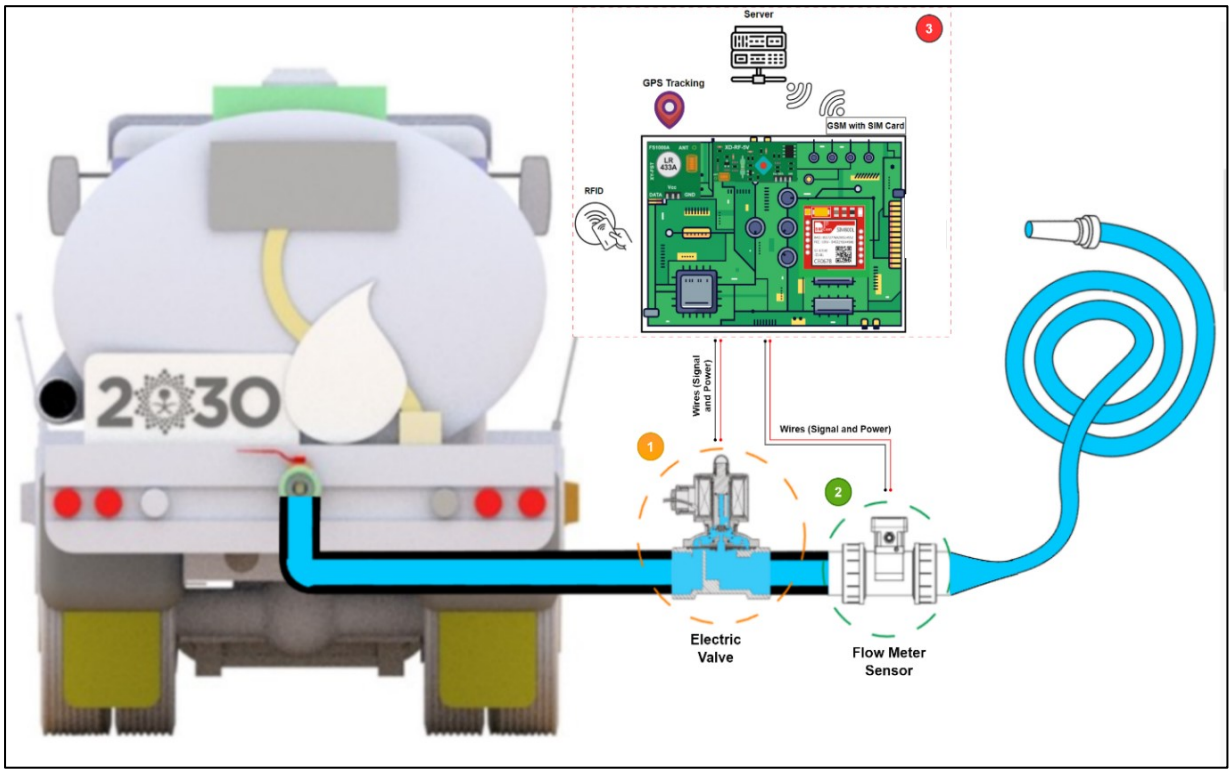


Image 1: Water Dispensing System

The solenoid valve is used to switch on or off the flow of water in by means of electricity. It connects to the system control unit that controlling its on-off operation.

The flow-meter sensor is used to measure the water flow rate. The flow-meter sensor consists of a copper body, a water rotor, and a hall-effect sensor. When water flows through the rotor, rotor rolls, its speed changes with different rate of flow. And the hall-effect sensor outputs the corresponding pulse signal. It controlling the amount of dispensing water by returning a feedback signal to the control unit which controlling the valve.

The control unit shown in image 2 will be designed to monitor and control the operation of the system. The control unit will include a microcontroller, peripherals, power circuits, connectors, sensors, and actuators interfacing circuits. The microcontroller will receive inputs from the sensors and control the operation of the valve as described in next section.

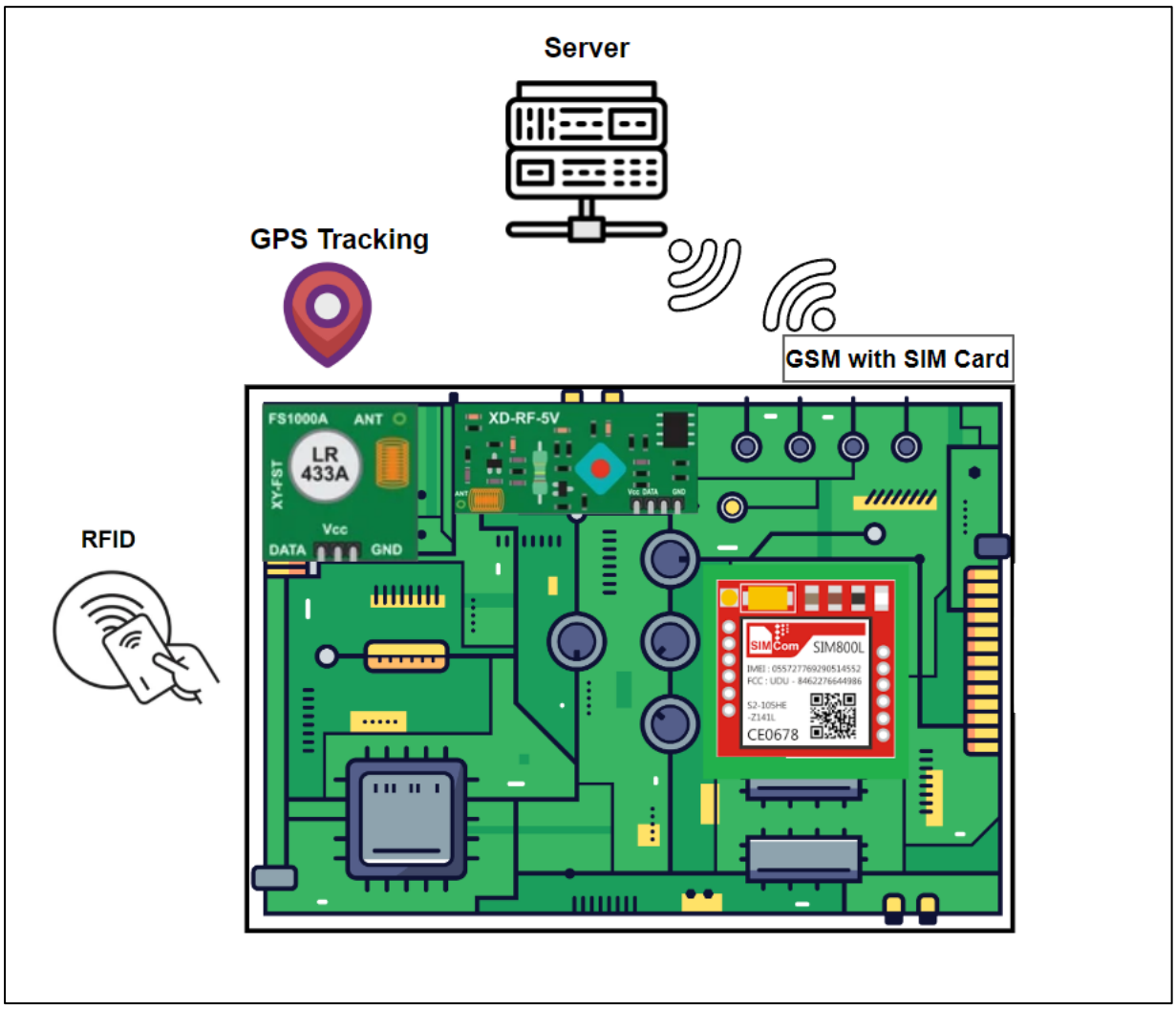


Image 2: System Control Unit

The system microcontroller is composed of both the primary MCU and secondary MCU. The main MCU will be an STM32 based chip. The STM32 microcontroller is a powerful and versatile microcontroller that has many advantages. It has a high processing power, low power consumption, real-time operating system support, extensive peripheral support, large memory capacity, ease of programming, and cost-effectiveness and a wide range of peripherals, including UART, I2C, SPI, CAN, and USB. The STM32 will handle the whole system functions.

Another microcontroller will be used as secondary MCU. It is used for connection handling with the system server. It is an ideal candidate for managing connections to a system server through a GSM network. The GSM module acts as a modem, allowing the secondary MCU to communicate with the system server via the cellular network. It offloads connection handling from the primary MCU, allowing it to focus on the core functionality of the project. It connects with the main MCU using one of common protocols like SPI or CAN Bus and connected to a GMS module through its UART interface.

The interfacing circuits include circuits for the flow meter sensor and the valve. The output of the flow meter sensor is a pulse signal that is proportional to the flow rate of the water. A design of an interfacing circuit is needed to that can detect the pulses from the flow meter sensor and convert them into digital signals that can be read by the microcontroller.

The valve is controlled by a digital signal that is sent to the valve's control circuit. To interface the solenoid valve with the STM32 microcontroller. A valve interfacing circuit can be designed to generate a digital signal that can be sent to the valve's control circuit.

The interfacing circuit for the flow meter sensor can be designed using a Schmitt trigger buffer, a voltage level shifter, and a digital input pin of the STM32 microcontroller. The Schmitt trigger buffer is used to clean up the pulses from the flow meter sensor, while the voltage level shifter is used to convert the pulse signal from the flow meter sensor to the appropriate voltage level that can be read by the digital input pin of the STM32 microcontroller.

The interfacing circuit for the solenoid valve can be designed using a MOSFET switch, a diode, and a digital output pin of the STM32 microcontroller. The MOSFET switch is used to turn the solenoid valve on and off, while the diode is used to protect the MOSFET switch from voltage spikes that may be generated by the solenoid valve.

A GPS and a communication system will be used to locate the truck and to navigate to the water filling locations. The GPS system uses the GSM module for connection.

In addition, the control unit has a Radio Frequency Identification (RFID) system. It consists form a tag and reader. The reader has one or more antennas that emit radio waves and receive signals back from the RFID tag. Tags is a portable key that carries the user information. The RFID will permit only the authorized pre-defined user to use the system.

The system server will have the website. It will be developed using core.NET technology. Core.NET is a popular framework for building web applications, services, and APIs. It provides a powerful and flexible platform for building robust and scalable applications that can interact with various devices and systems, including microcontrollers. It provides a lightweight, high-performance, and modular HTTP request pipeline feature. The control unit will create a real-time communication with the website to fetch the detailed information about the filling process. The website has a use graphical interface page with a level of permission for each user.

Solution description

The objective of this solution is to track location and water dispensing process for water trucks. This solution enables the admin to track water truck fleets through a GPS mounted on the trucks and the software provided. In addition, track & allow water dispensing process through a two-authentication system Geofence and RFID reader on the truck.

An RFID tag will be distributed to clients, in which the company will store clients Tag data, geofence location, and information on the software provided. Another authentication method is, Geofence and OTP to the linked phone will be sent in case tag is not available on premises, in which can be inserted on the keypad for client dispensing confirmation. Once the truck is set to a certain route and reaches

its destination. The geofence authentication will confirm the area is correct through the server database or onboard memory database in case of no coverage areas or connectivity issues.

The second authentication will be the client tag. Once provided to the driver reader will confirm tag and the screen on the truck will show client info and how much liters to be dispensed for this unit and then fill it automatically. In case tag is not available, OTP can be used to confirm operation. Afterwards, truck driver pushes the button to start dispensing. Once the client unit has been refilled a receipt of the operation is printed through the thermal printer on truck then the device will send the data of the process to the server so the admin can see the data live or through records through software provided.

Component List:

Hardware:

- **MCU:** analyze the input and make a decision on the output.
- **GPS tracker:** sends the location every specific amount of time.
- **GSM with SIM card:** providing internet access to the control panel.
- **Thermal printer:** print a receipt for the client.
- **Screen:** to display client's information. Name and liters to fill.
- **Keypad:** typing one time password.
- **Button:** after success verification pushing the button will start filling.
- **RFID reader:** reads clients tag information.
- **RFID tag:** held by the client to puts it near the RFID when he requests tank refill
- **Flow meter:** measuring the amount of water flowing through the hose by liters.
- **Electric valve:** allow the liquid to flow by command.
- **TDS sensor:** measures the number of total dissolved solids like salts, minerals, and metals in the water.

Technology description

This solution takes a box style mounted on the water truck powered with 24V and it has an integrated RFID reader to read clients RFID tag and an integrated keypad to type one time password and finally, an integrated screen to show clients information such as name and how much liters to fill using flow meter located in the middle of the hose, if the client verified and button pushed electric valve will open to start filling the tank, the filling process will be sent automatically to the admin using the server to match the information he got to the information he has, If it is a water truck the TDS sensor will be placed in the tank or in the hose to measure the water quality

Note:

According to the battery and generator of the trucks output voltage is 24V, it is possible to connect DC-DC converter to input to step down the voltage for the circuit.

Design Stages

Component selection

The component selection process for a smart water dispensing system is a critical aspect of the system design process. The components selected should be able to meet the system's requirements and operate efficiently and effectively. The following is a detailed discussion on the selection of the electric valve, water flow meter sensor, pump, and control unit for a smart water dispensing system.

MCU

The system microcontroller is composed of both the primary MCU and secondary MCU. The main MCU will be an STM32 based chip. The STM32 microcontroller is a powerful and versatile microcontroller that has many advantages. It has a high processing power, low power consumption, real-time operating system support, extensive peripheral support, large memory capacity, ease of programming, and cost-effectiveness and a wide range of peripherals, including UART, I2C, SPI, CAN, and USB. The STM32 will handle the whole system functions.

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Features:

- Power Supply:
    - Voltage: 9-36V DC
- GPS Navigation:
    - The antennas are both internal or external
    - Receiver: uBlox NEO 6M engine
    - Sensitivity: -162dBm
    - Navigation Update: 1Sec
    - TTFF: Cold Starts: 29s/Aided Starts: <1S/ Hot Starts: <1S
- GSM:
    - The antennas are both internal or external
    - Modem: QUECTEL M35 /Quad Bands:850/900/1800/1900MHz
- Radio Frequency:
    - Wi-Fi module (Optional)
    - hotspot module (Optional)
    - Working frequency: 433MHz.
    - Code: 1527/2240 study
    - Working Voltage: 12V (27A 12V dry battery)
    - Control Distance: 20-50M
- Environmental:
    - Working Temp. -20℃ ~ +75℃
    - Humidity: 20-95%
- Internal Memory
    - Use 16M flash to save 96000 points all status
- Voice Monitor
    - Two-way voice monitoring (Optional).

- Input
  - Ignition on/off detection
  - Door open/close detection
  - SOS (Panic) detection (Optional)
  - AD input (Temperature sensor) (Optional)
  - multiple analogue/digital inputs
- Output
  - Central locking control
  - multiple analogue/digital ports

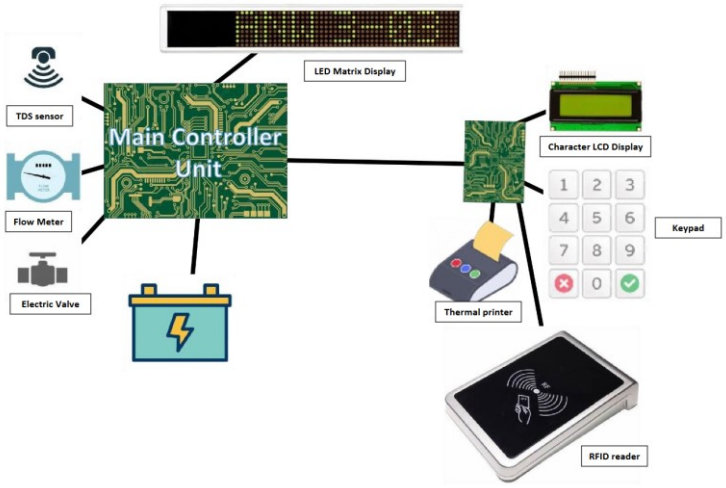


Image 3: Illustrative image of MCU and Connected modules

Electric Valve

The valve should be able to operate efficiently, have low power consumption, and be reliable. The valve should also be compatible with the control unit and the pump. The control valve should be selected to control a flow rate from 500 to 1000 liters per second. It should be made of durable and corrosion-resistant materials. The valve has to be compatible with the system voltage range.



Image 4: Illustrative image of Electric Solenoid Valve

Water Flow Meter Sensor

A water flow meter sensor is used to measure the flow of water in the system. One of most critical points of the flow meter sensor is the nominal flow rate. It should be able to measure the system maximum flow rate which is about 1000 liters per second. Moreover, the operating temperature is another critical parameter that should be considered when selecting a water flow meter sensor. The sensor should be able to operate effectively under high outside temperatures. This is particularly important for systems that will be used in hot environments, as the temperature can affect the accuracy of the sensor's readings.

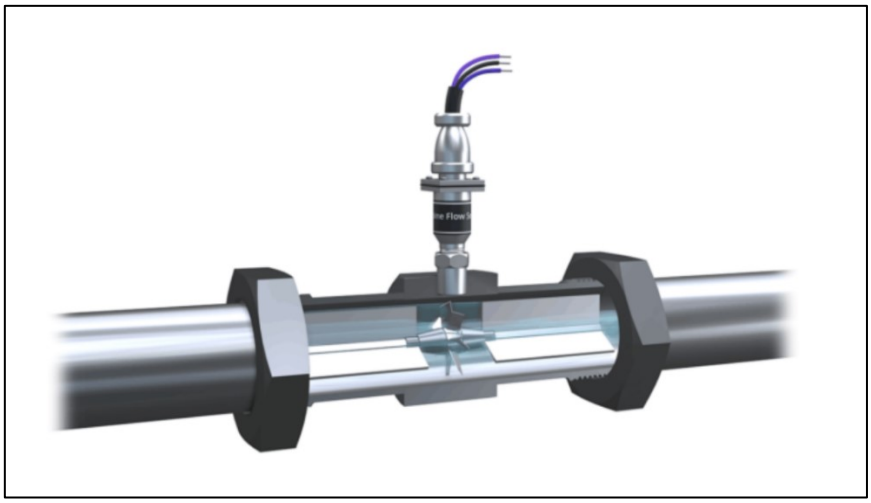


Image 5: Illustrative image of Water Flow Meter Sensor

GSM Module

The GSM module should be compatible with the system's control unit and the cellular network in the area to ensure reliable communication between the device and the system's control unit. It must offer a GPS feature with high accuracy and sensitivity. This is important for accurate geofencing and GPS tracking. Higher-end GPS modules generally offer better accuracy and sensitivity, but they can also be more expensive. It is important to balance the cost with the required level of accuracy for the specific application. It is important to select a GPS module that has the appropriate level of tracking accuracy and frequency.



Image 6: Example of GSM Module

RFID Module

The RFID module should be compatible with the system's control unit to ensure seamless integration and communication. Additionally, the type of RFID tags being used, such as passive or active tags. The read range will determine how far away an RFID tag can be read by the module. A longer read range will allow for greater flexibility in the placement of the RFID tags, but may also increase the risk of unintended reads. Moreover, the read speed will determine how quickly the RFID module can read the information from the RFID tag. The module should have secure authentication protocols to prevent unauthorized access to the system. This is especially important in a water dispensing system where security is essential to prevent tampering or unauthorized access to the water supply.



Image 7: Illustrative image of RFID Module

GPS tracker

The reason for a GPS tracking device is to track the truck location to ensure the admin if the driver goes in the right way and sees the reason if he is late.



- Live & Near-Real Time Location Tracking Using the Internet
- Advance Secured Technology & GPS Coordinates
- Power Loss Detection & Alert
- Discrete & Hidden Installation of the Tracking Device
- Instant Alerts & Notifications via Monitoring Dashboard, Emails & SMS
- 24/7 Monitoring
- Customization & Tailor-Made Solutions
- Wi-Fi module, hotspot module (Both are Optional)
- 4G module
- 2x RS232
- Setup geofence from website
- Send/receive commands
- Can Control Vehicle on/off remotely, and red measurements from vehicle

Thermal printer

A thermal printer is a type of printer that uses a heated thermal printhead pressed against a medium to produce black text or images.



Types
Direct Thermal Printing
Thermal Transfer Printing

1. Multipurpose: Print logo, language, number, barcode

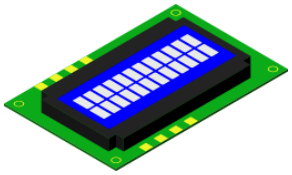
2. CE certificated
3. Fashionable, delicate and elegant
4. Low noise direct thermal printing method
5. Various image/curves/characters printing function
6. Paper out detection, automatic memory printing function
7. Serial (RS232, TTL)/Parallel interface
8. High speed and low power consumption optional
9. Support low-voltage system

Displays



Character LCDs, or alphanumeric displays, can display letters (capital or non-capital), numbers, and punctuation. You can find this type of display in products such as vending machines, 3D printers, data loggers, etc. An LED matrix or LED display is a large, low-resolution form of dot-matrix display

character LCD



displays are available from 8x2, 12x2, 16x1, 16x2, 16x4, 20x2, 20x4, 24x2 through to 40x4 formats with 5x8 dot matrix characters.

Types

LED matrix

Note:

Based on the client request, to use an LCD screen instead of the dot matrix in order to be able to present animation and further details. This request is under consideration and being examined to determine if it will affect the schedule of delivery throughout the process.

Keypad



A keypad is a set of buttons or keys bearing digits, symbols and/or alphabetical letters placed in order on a pad, which can be used as an efficient input device.

Types

- Capacitive Keypad
- Scissor-Switch Keypad
- Dome-Switch Keypads
- Membrane Keypads

Button

A button/switch or knob that when pushed operates something especially by closing an electric circuit.

Types



- Flush push button switch
- Extended push button switch
- Mushroom push button switch
- Emergency stop push button switch
- Illuminated push button switch
- Double push button switch

Control Unit PCB Design

The Control Unit PCB (Printed Circuit Board) consists of several layers of conductive materials, including copper traces and pads that connect the different components of the system. These layers are sandwiched between insulating layers, which provide electrical isolation and mechanical support.

The Control Unit PCB design must be carefully engineered to meet its specific requirements. This includes the size and shape of the board, the number and type of components it will support, and the power requirements of the system. The design process involves several stages, including component selection, schematic capture, layout design, verification, and Gerber files.

Schematic capture involves creating a diagram that represents the electronic system's logical structure, including the components and their connections as shown in image 7.

The pictures below outline the first-level and initial design of the circuit for the system.

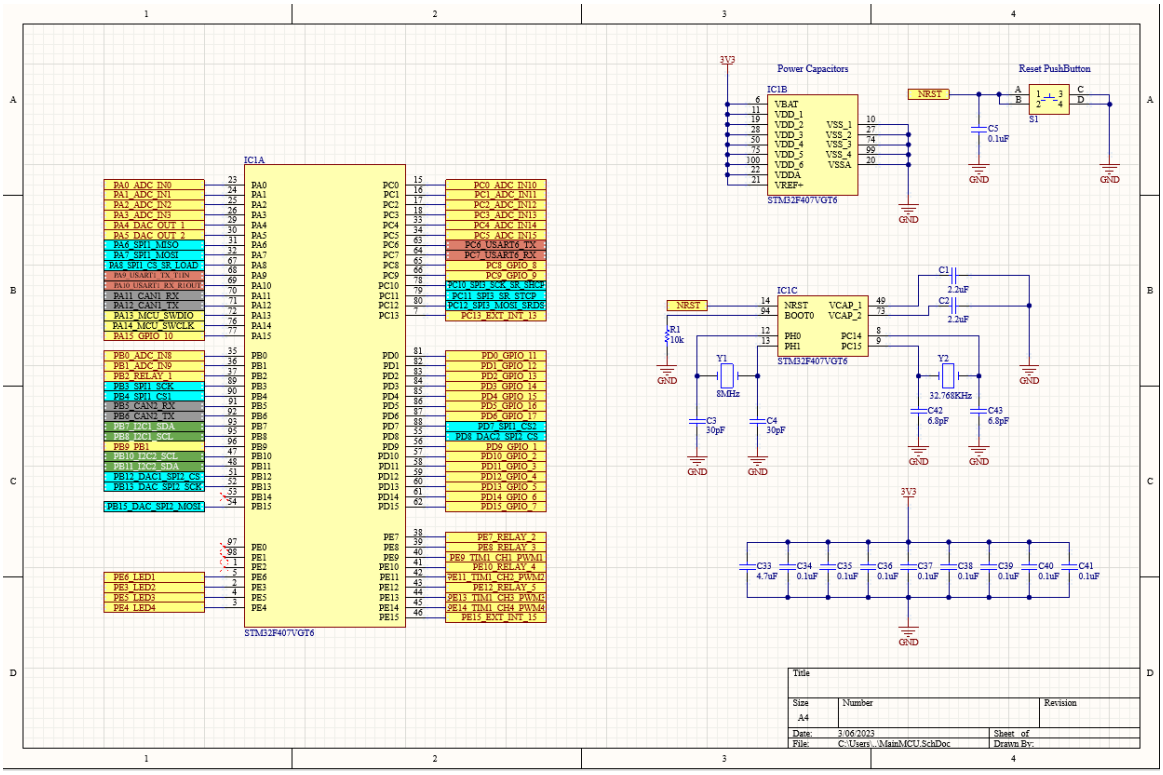


Image 8: PCB Schematic

The layout design stage involves taking the schematic and translating it into a physical design, placing the components on the PCB and routing the necessary connections between them as shown in images 8 and 9. This stage requires careful attention to detail, as any mistakes in component placement or routing can result in system malfunctions or failure.

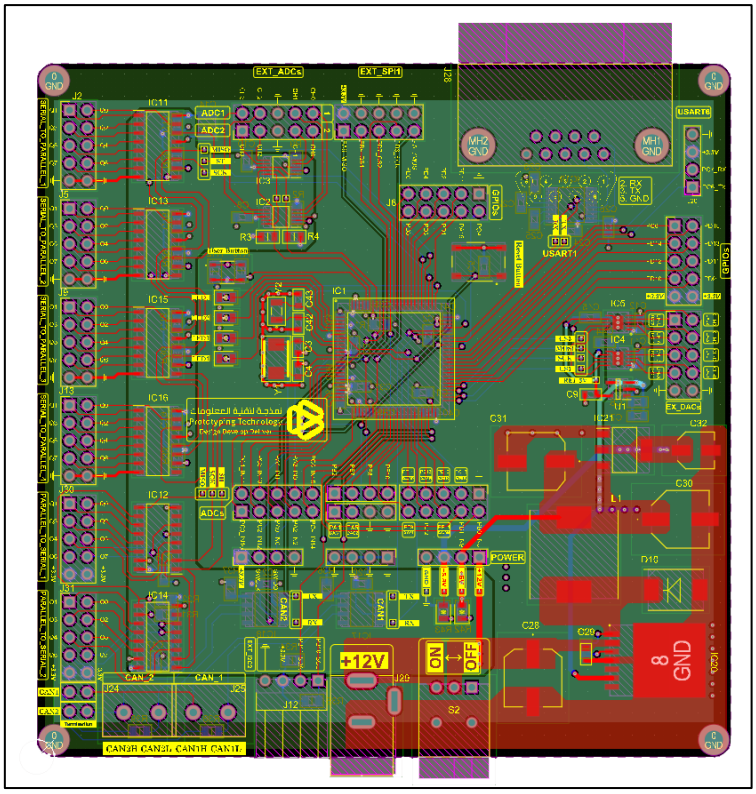


Image 9: Board Layout

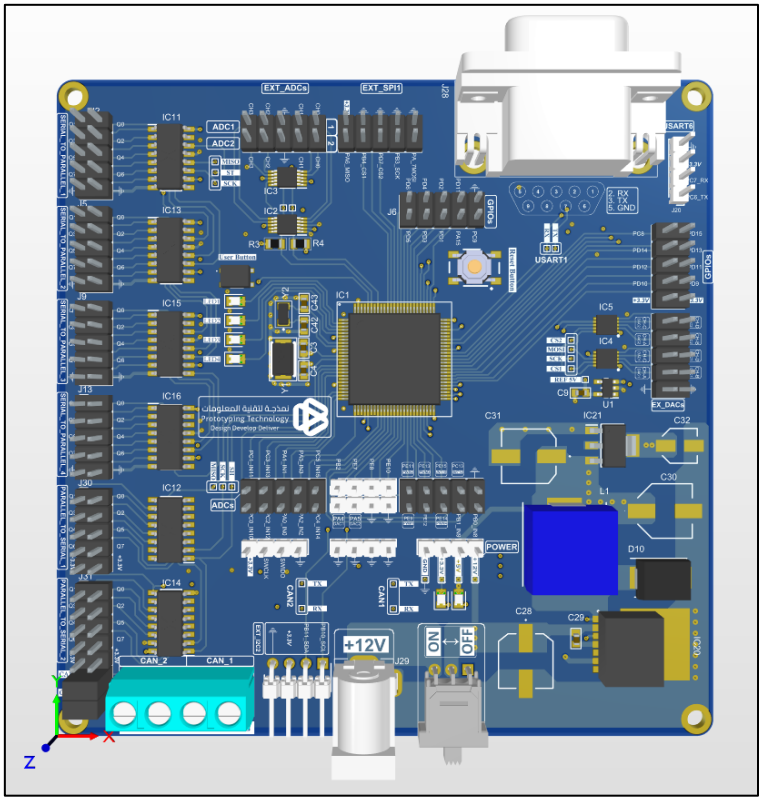


Image 10: Board Layout 3D View

Verification involves testing the completed design to ensure that it meets the system requirements and performs as expected. This may involve using simulation tools to test the design before building

the physical PCB, as well as physical testing of the completed PCB to ensure that it meets the desired performance metrics.

After the PCB layout is complete, we will generate the Gerber files required for manufacturing. The Gerber files are used by the manufacturer to create the PCB. The Gerber files will include information about the copper traces, drill holes, and other details required for the PCB.

Control Unit Software Development

Control unit software development is a critical aspect of creating electronic devices and systems that require real-time control and monitoring. The control unit software serves as the interface between the hardware and the user, enabling them to interact with the system and control its behavior. The software development has many stages, including architecture design, coding and implementation, and testing.

Once the requirements have been identified, the next step is to design the software architecture. See image 10. This involves defining the structure and organization of the software and deciding on the programming language, platform, and tools to be used.

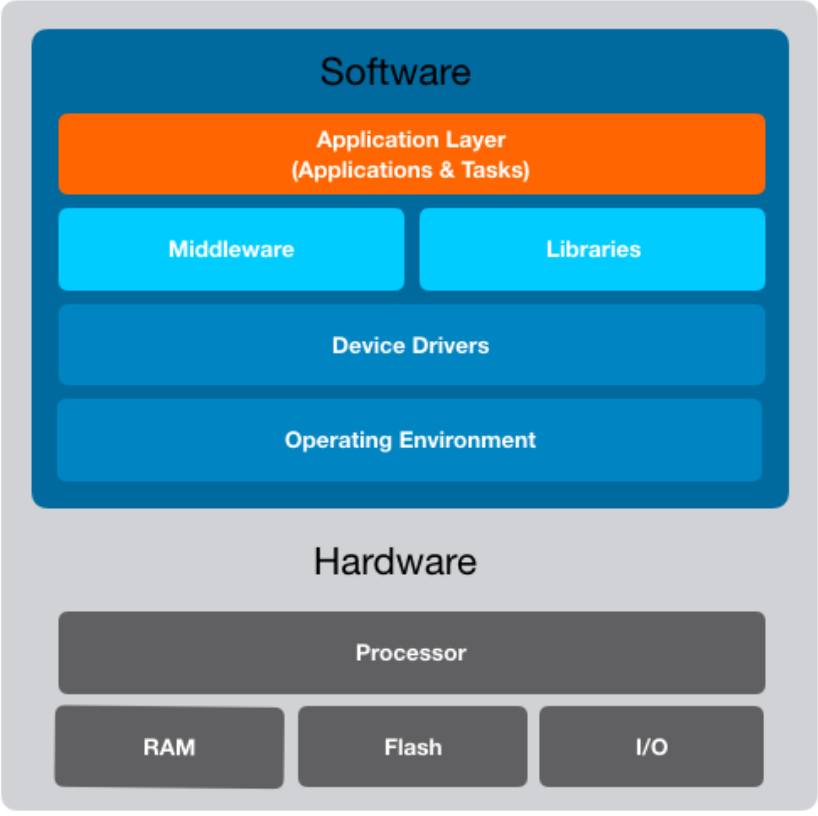


Image 11: Software Architecture

The architecture design considers the system's scalability, maintainability, and flexibility for future updates and modifications.

The coding and implementation phase is where the actual software is created as show in image 11.

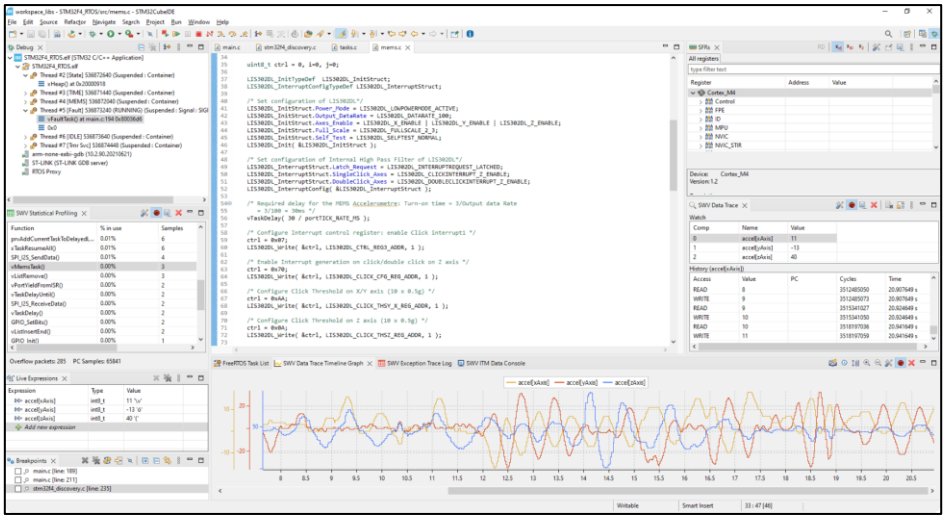


Image 12: Software IDE

The software is designed to interact with the hardware components, handling inputs and outputs in real-time. The code must be optimized for speed and efficiency to ensure that the system responds quickly to user requests and environmental changes.

Once the software has been developed, it must be thoroughly tested to ensure that it functions correctly and meets all the requirements. This involves performing unit tests, integration tests, and system tests to verify that the software operates as expected. Any issues or bugs discovered during testing must be addressed and corrected before the software can be released.

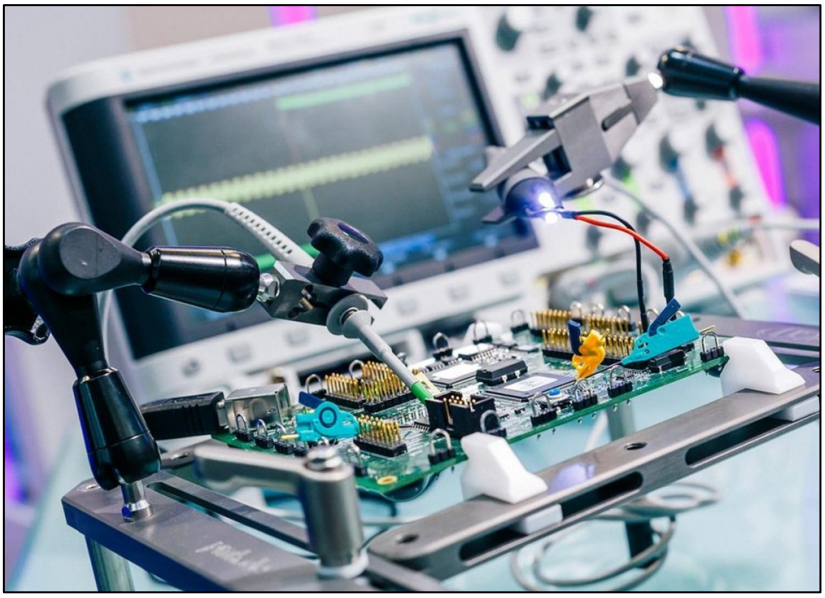


Image 13: Control Unit Testing and Troubleshooting

Build The System Mechanical Frame

The mechanical frame of a system is a critical component that provides support and structure to the various mechanical and electrical components of the system. It is the foundation on which the rest of the system is built, and as such, it is essential to get the frame design and build right.

Design and engineering the first step in building the system mechanical frame is to design and engineer the frame. This involves determining the dimensions, shape, and materials required for the

frame based on the system's specifications and requirements. The design must consider factors such as weight, size, and durability.

The mechanical frame design includes the system board enclosures design as shown in image 12, and the system frame that carries the system components.

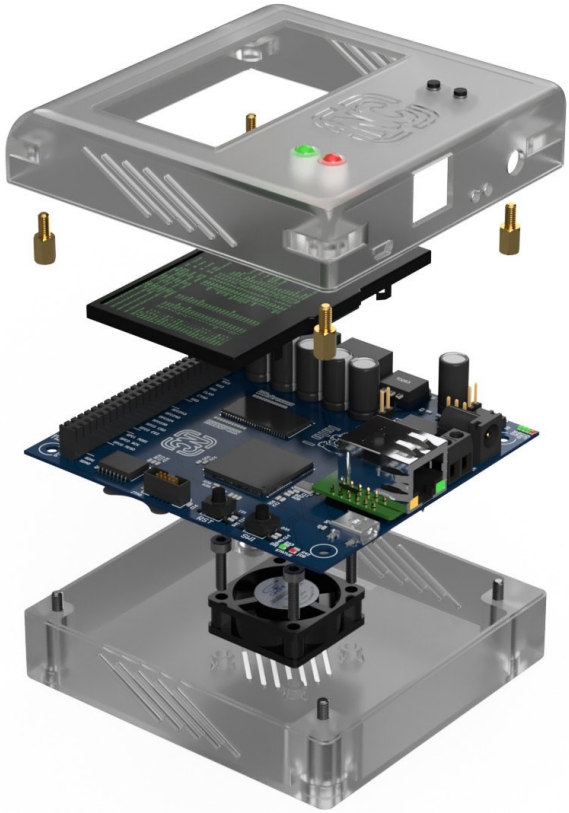


Image 14: Board Enclosure Design

Once the design is finalized, the next step is to select the appropriate materials for the frame. The materials used will depend on the design specifications and the intended use of the system.

The next step is to manufacture and assemble the frame. This involves cutting and shaping the materials, and then welding or bolting them together to create the frame. The assembly process must be done carefully and precisely to ensure that the frame is stable and strong.

When the frame is assembled, the next step is to mount the various components of the system onto the frame. This may include mounting the electrical components, actuators, sensors, or other mechanical components onto the frame. The mounting must be done securely to ensure that the components do not move or vibrate during use.

After the components are mounted onto the frame, the system must be tested and validated to ensure that it operates correctly and meets the design specifications. This involves testing the system under various conditions to ensure that it is stable, durable, and reliable.

Finally, after testing and validation, the system mechanical frame may need to be refined and improved to optimize its performance or to address any issues that were discovered during testing. This may involve making modifications to the design, materials, or assembly process.

Build The System Prototype

Building a system prototype is an essential step in the process of developing a new product or service. It involves creating a preliminary version of the system that can be tested and evaluated to identify any potential issues or areas for improvement. The prototype serves as a tangible representation of the final product, allowing developers to visualize and refine the design before investing resources into full-scale production.

Once the components have been identified, they can be assembled using a variety of techniques and tools. This may involve using specialized equipment and tools. The assembly process should be carefully planned and executed to ensure that all components are properly aligned and that the final product meets the desired specifications.

Once the system has been assembled, it should undergo thorough testing and evaluation to ensure that it meets the intended functionality and performance requirements. This may involve functional testing, stress testing, or other evaluation techniques to ensure that the system is working as intended. Based on the results of these tests, the design can be refined. And any necessary adjustments can be done.

System Features

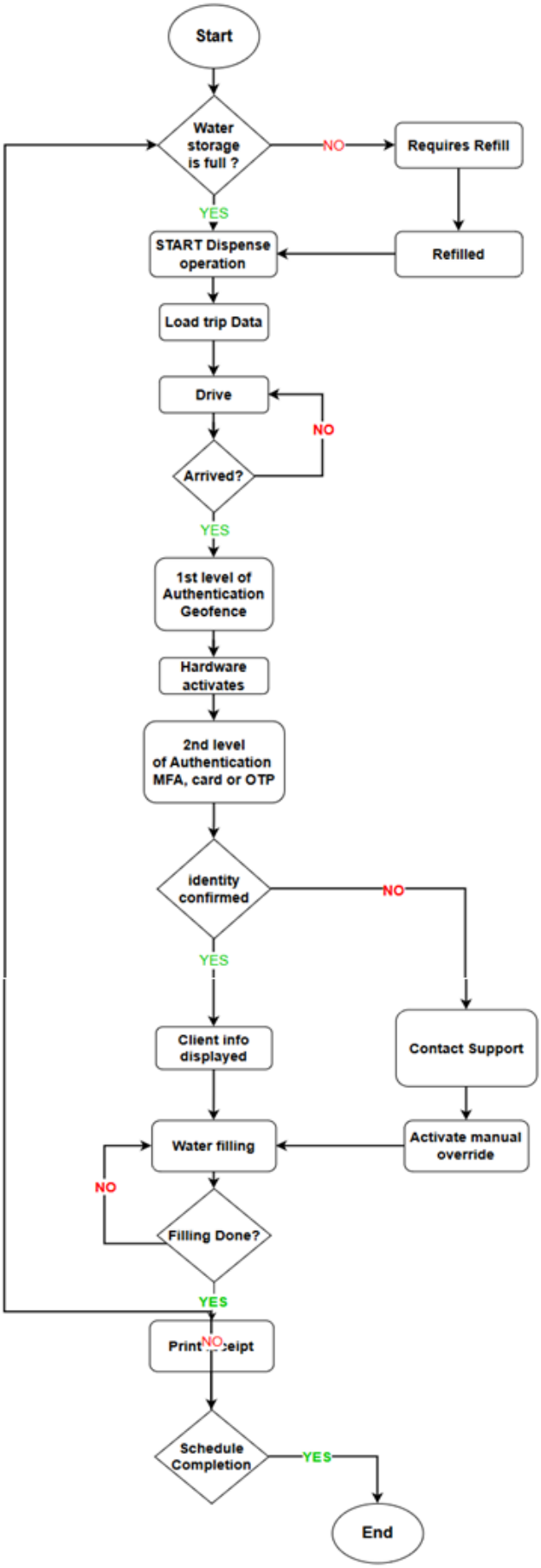
The smart water dispensing system provides a unique feature that are concluded as follows:

- It provides a multi-factor authentication (MFA) method that requires the user to provide more than one verification factors to gain access to the system.
- It is a clean and safe water in areas with limited access to water.
- The services will still run even in offline status (not in connection with the server)
- It uses the Geo-fence technology that used to alert the user with the truck is in the defined area.
- The system is secure and operates only on the pre-defined locations.
- The system provides the predefined amount of the water for each client.

It gives the users permissions and level of control as defined in the system database.

2nd: Software

Initial Flowchart based on the Hardware Function


Start
Water storage is full ?
NO
Requires Refill
YES
START Dispense operation
Refilled
Load trip Data
Drive
NO
Arrived?
YES
1st level of Authentication Geofence
Hardware activates
2nd level of Authentication MFA, card or OTP
identity confirmed
NO
YES
Client info displayed
Contact Support
Water filling
Activate manual override
NO
Filling Done?
YES
Print Receipt
NO
Schedule Completion
YES
End

Software:

- Add client details,
- Link tag to client
- Allocate the amount that should be dispensed.
- Track trucks progress
- Check water purity
- Live records, daily, weekly, and monthly of operations performed

Case overview

An organization that has a water truck fleet that dispenses to its clients through a database. This organization faces issues in calculating and managing the amount of time it takes to refill for a set of clients and keeping records of how much water dispensed per client. For example, it should take five hours and 1000 liters to be dispensed at the end of the route. However, the truck returns with 200 liters remaining and all clients confirmed that they got their water. This causes problems and without a system that shows the steps it is hard for the employee, employer, and client to track the issue. This is where our Water and Fleet management solution comes into places through the following components integrated together.

Service simulation

When the truck goes to client's location the GPS device will send location every specific amount of time using GSM to the admin. After the truck arrives to the client's location the client will put the RFID tag near the RFID reader and typing the password on the integrated keypad, the screen will display tag information such as the client's name and liters to fill in the tank if the reader reads the tag properly the solenoid valve will open to start filling the tank automatically according to saved client data after filling is done the device will send the water filling process data using GSM to the admin and finally, print receipt using thermal printer.



FMEA – failure mode and effects analysis

When the client is in an area outside the network coverage a problem may happen, that issue may happen when the device cannot reach the internet because of this area, the possible solution is providing the device with SD card to save the client data in it.

The second scenario is when the truck is water truck an issue may appear in the TDS sensor, the issue is the accumulation of impurities the TDS sensor, the accumulation of impurities issue makes the sensor measuring not accurate, the solution of this issue is cleaning the TDS sensor more often.

Risk & Mitigation Plan

Intro

In this project risk management and its associated mitigation efforts is differentiated in two areas:

1. Risks related to the project's execution: The risks grouped in this area range from technical risks which can be directly related to the project progress, e.g. management issues, installation, etc.
2. Risks due to external factors (External Risks): The risks grouped in this area are risks that cannot be directly influenced. Typically, these risks can be for example: proposed policies, procedures and standards.

Risk management includes up-front planning of how risks will be mitigated and managed once identified. Therefore, risk mitigation strategies and specific action plans are taken care off in the project via both a dedicated task and a dedicated risk manager. Typically risk mitigation plans should characterize:

- The root causes of risks that have been identified and quantified in earlier phases of the risk management process;
- Evaluate risk interactions and common causes;
- Identify alternative mitigation strategies, methods, and tools for each major risk;
- Assess and prioritize mitigation alternatives;
- Select and commit the resources required for specific risk mitigation alternatives;
- Communicate planning results to all project participants for implementation

Risk Mitigation Procedures

Best practices in this area require that the known and perceived risks will be prioritized according to the degree and likelihood of the disadvantageous results that are anticipated to take place. Followed by a thorough analyses and documentation of all such risks according to their levels of priority in a form known as the risk mitigation plan. After which, the development and integration of the corresponding risk mitigation strategies follows.

Generally, three key questions can be posed for risk mitigation:

- What can be done and what options are available?
- What are the trade-offs in terms of all costs, benefits, and risks among the available options?
- What are the impacts of current decisions on future options?

An understanding of these three questions is essential to risk mitigation and risk management planning for project execution. Question 1 addresses the available risk response options, an understanding of questions 2 and 3 is necessary for risk planning because they determine the impact of both the immediate mitigation decisions and the flexibility of risk mitigation and planning on future events.

Process

The team will work together in order to ensure that risks are actively identified, analyzed, and managed throughout the life of the project. Risks will be identified as early as possible in the project so as to minimize their impact.

Risk Identification

Risk identification will involve the members and will include an evaluation of external factors that might hamper or endanger the main goal of the project. Careful attention will be given to external factors financial feasibility, etc. An Excel based Risk Management Tool will be generated and updated as needed. The qualitative Risk Analysis as described below is part of the tool.

Risk Analysis

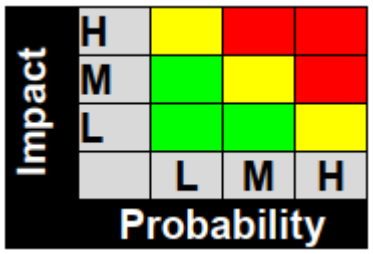
All risks identified will be assessed to identify the possible effect on the project objective. Qualification (ranking) will be used to determine which risks are the top risks to pursue and respond to and which risks can be ignored.

Qualitative Risk Analysis

The probability and impact of occurrence for each identified risk will be assessed by the Manager, with input from the consortium members:

Probability

- High – Greater than 75% probability of occurrence
- Medium – Between 25% and 75% probability of occurrence
- Low – Below 25% probability of occurrence



Impact

- High – Risk that has the potential to greatly impact project objective
- Medium – Risk that has the potential to slightly impact the project objective
- Low – Risk that has relatively little impact on the project objective

Risks that fall within the RED and YELLOW zones will have risk response. Planning which may include a risk mitigation and a risk contingency plan.

Quantitative Risk Analysis

Based on the qualitative risk analysis, the risks will be prioritized into a “Top 10 Risk List”.

Risk Response Planning

Each major risk (those falling in the Red & Yellow zones) will be assigned to a team member. For each major risk, one of the following approaches will be selected to address it:

- Avoid – eliminate the threat by eliminating the cause

- Mitigate – Identify ways to reduce the probability or the impact of the risk
- Accept – Nothing will be done
- Transfer – Make another party responsible for the risk (buy insurance, outsourcing, etc.)

For each risk that will be mitigated, the project manager will identify ways to prevent the risk from occurring or reduce its impact or probability of occurring. For each major risk that is to be mitigated or that is accepted, a course of action will be outlined in the event that the risk does materialize in order to minimize its impact. This course of action should however be reasonably and within the overall scope of the project.

Risk Monitoring, Controlling and Reporting

The level of risk will be tracked, monitored and reported throughout the project lifecycle. A "Top 10 Risk List" will be maintained by the Risk Manager and will be reported as a component of the project status reporting process for this project.

Conclusion

This document contains all of the information needs to build this solution from zero including components, process and specifications and all of the details require even some of the possible failure scenarios to provide their effects.

This solution helps organizations to Reduce manipulation and delay by tracking their fleet locations easily by the administrator and the liquid filling process if the liquid is water, diesel or water just by providing the correct components this solution will suit any type of the liquid.